

Q1. Write a Selenium WebDriver program in Java to switch to a frame (by index or name), enter text into a textbox inside the frame, and then return to the main page.

Input

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class FrameExample {
    public static void main(String[] args) {

        WebDriver driver = new ChromeDriver();

        driver.get("https://the-internet.herokuapp.com/iframe");

        // Switch to frame using index
        driver.switchTo().frame(0);

        // Enter text inside the frame
        driver.findElement(By.id("tinymce")).clear();
        driver.findElement(By.id("tinymce")).sendKeys("Hello Selenium!");

        // Return to main page
        driver.switchTo().defaultContent();

        driver.quit();
    }
}
```

Q2. Explain the difference between handling multiple windows and multiple frames in Selenium. Give examples.

Multiple Windows	Multiple Frames
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A new browser window or tab opens.	A frame is a section inside the same webpage.
Selenium uses <code>getWindowHandles()</code> and <code>switchTo().window()</code> .	Selenium uses <code>switchTo().frame()</code> .
Each window has a unique Window ID.	Frames are identified by index, name, or <code>WebElement</code> .
Used for pop-up windows, payment pages, login windows, etc.	Used for embedded content like ads, editors, maps, videos, etc.

Example of Multiple Windows

- Clicking “**Login with Google**” opens a new browser window.
- Selenium switches using Window Handle.

Example of Multiple Frames

- A webpage contains an embedded text editor or advertisement inside an `iframe`.
- Selenium switches using `driver.switchTo().frame()`.

Q3. Write a Selenium WebDriver script to:

- Open a website
- Capture a screenshot of the homepage
- Save it in the local project folder as `homepage.png`

```
import org.openqa.selenium.OutputType;
```

```
import org.openqa.selenium.TakesScreenshot;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.chrome.ChromeDriver;

import java.io.File;
import org.openqa.selenium.io.FileHandler;

public class ScreenshotExample {

    public static void main(String[] args) throws Exception {

        WebDriver driver = new ChromeDriver();

        driver.get("https://www.google.com");

        File src = ((TakesScreenshot) driver).getScreenshotAs(OutputType.FILE);

        File dest = new File("homepage.png");

        FileHandler.copy(src, dest);

        System.out.println("Screenshot Saved");

        driver.quit();
    }
}
```

```
}
```

Q4. What is a broken link? Write a short Selenium program outline to find and print all broken links on a webpage.

Theory

A broken link is a hyperlink that does not work because the webpage is missing or unavailable. It usually returns HTTP errors like 404 (Not Found) or 500 (Server Error).

```
import org.openqa.selenium.By;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.WebElement;

import org.openqa.selenium.chrome.ChromeDriver;


import java.net.HttpURLConnection;

import java.net.URL;

import java.util.List;


public class BrokenLinks {

    public static void main(String[] args) throws Exception {

        WebDriver driver = new ChromeDriver();

        driver.get("https://example.com");

        List<WebElement> links = driver.findElements(By.tagName("a"));
```

```
for (WebElement link : links) {

    String url = link.getAttribute("href");

    if (url == null || url.isEmpty())

        continue;

    HttpURLConnection connection =

        (HttpURLConnection) new URL(url).openConnection();

    connection.setRequestMethod("HEAD");
    connection.connect();

    int code = connection.getResponseCode();

    if (code >= 400) {

        System.out.println(url + " -> Broken Link");

    } else {

        System.out.println(url + " -> Valid Link");

    }

}
```

```
driver.quit();
```

```
}
```

```
}
```